- *Self-described and discoverable*—Data should be described, and this description should be an integral part of the data product. The data product should be able to register itself in the data mesh ecosystem, and, as a result, should be discoverable.
- *Addressable*—It should have a unique address (e.g., in the form of a URI address), so that it can be referred to and relationships between data products can be created.
- *Interoperable*—A data product should be made according to predefined standards, concerned with the form of data sharing, standardized formats, vocabularies, terminology or identifiers, security, and trustworthiness. A data product should meet the established and declared service-level agreement (SLA), and should enable controlled access, thus ensuring data security from both perspectives—intellectual property and regulations such as the European Union's General Data Protection Regulation (GDPR).

DATA PRODUCT AS AN AUTONOMOUS COMPONENT

While fulfilling the previously established conditions, a data product should at the same time constitute an autonomous component, so that it can be independently developed by the team responsible for it. From the point of view of a technical solution, we can see a data product as an analog of a microservice in the data world. In addition to making data available, the data product embeds code related to data transformation, cleaning, or harmonization. It also exposes interfaces for automatic integration with the data mesh environment and platform by providing, among other things, the following:

- *Input logic and output ports*—Form and protocols used to ingest from the source and expose data to consumers
- *Operational metrics*—Number of users, throughput, amount of data fetched, etc.
- *Data quality reports*—Quantity of incomplete data, format incompatibilities, statistical measures of outliers, etc.
- *Metadata*—Specification and description of the data schema, domain description, business ownership, etc.